

Intelligent News Credibility Analysis

& Agentic Misinformation Monitoring

Milestone 2 — End-Semester Technical Report

From Predictive NLP to Autonomous Fact-Checking

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Core Stack: LangGraph • ChromaDB/FAISS • FastAPI • Streamlit • TF-IDF + Linear SVM • NewsAPI • Open-Source LLMs

Abstract

This report presents the complete design, architecture, and phased implementation plan for Milestone 2 of the Intelligent News Credibility Analysis project (Project 11). Building on the classical Machine Learning pipeline of Milestone 1—which achieved **99% F1-score** using a Linear SVM classifier trained on the ISOT Fake News Dataset—this milestone evolves the system into a fully **hybrid, agentic misinformation monitoring platform**. The architecture integrates five major intelligence layers: a content extraction layer, the Milestone 1 ML classifier as a preliminary signal, a Static Retrieval-Augmented Generation (RAG) subsystem backed by ChromaDB using the LIAR dataset, a Dynamic RAG layer via NewsAPI for real-time evidence, and an LLM-based reasoning engine orchestrated through a LangGraph agent workflow. The system accepts news articles as URLs or raw text and produces structured, explainable credibility reports containing a verdict, confidence score, identified risk factors, retrieved evidence, and source links. A FastAPI backend and Streamlit frontend complete the full-stack deployment, with mandatory public hosting on Streamlit Community Cloud or Hugging Face Spaces.

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1 Introduction

The rapid proliferation of misinformation on digital media platforms has created a pressing societal challenge. Manual fact-checking is inherently reactive, slow, and unable to scale to the volume of online content produced daily. Automated credibility assessment systems offer a proactive and scalable alternative, but classical approaches suffer from a critical limitation: they produce a binary prediction without context, evidence, or explanation. End users—journalists, researchers, and the general public—need to understand *why* an article is flagged, not merely *that* it is flagged.

Milestone 1 addressed the classification problem effectively, achieving near-perfect accuracy through disciplined feature engineering and model selection. Milestone 2 builds directly upon this foundation by introducing three capabilities that were absent in the first phase: **explainability** through evidence-backed reasoning, **recency** through live news retrieval, and **agentic orchestration** through a structured LangGraph workflow.

1.1 Motivation

Explainability: Users require evidence-backed reasoning rather than black-box scores.

Recency: Static training data becomes stale; misinformation evolves rapidly across news cycles.

Cross-source Verification: A single model cannot capture the diversity of misinformation patterns across outlets and regions.

Agentic Reasoning: Structured, multi-step workflows enable systematic and reproducible analysis with explicit state tracking.

2 Milestone 1 Recap

2.1 System Summary

Milestone 1 delivered a fully functional, deployment-ready ML-based fake news classifier using classical NLP and supervised learning, strictly excluding Generative AI per project constraints. The pipeline is summarised in Table 1.

Table 1: Milestone 1 Pipeline Summary

Component	Detail
Dataset	ISOT Fake News Dataset (Fake.csv + True.csv); title and text merged; shuffled and de-duplicated
Train/Test Split	80% training, 20% testing
Preprocessing	Lowercasing; URL/number/punctuation removal; negation-preserving stop-word removal; lemmatisation; duplicate elimination
Vectorisation	TF-IDF with <code>ngram_range=(1,2)</code> , <code>max_features=15000</code> , <code>min_df=5</code> , <code>max_df=0.9</code>
Models Evaluated	Logistic Regression, Multinomial Naive Bayes, Linear SVM
Best Model	Linear SVM (F1 = 0.9942)

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Hyperparameter Tuning	3-fold GridSearchCV on regularisation C ; optimal $C = 10$
Final Accuracy	≈99% on unseen test data
Saved Artefact	milestone1/model/news credibility model.pkl

2.2 Model Comparison and Final Performance

Table 2: Baseline Model Comparison

Model	Accuracy	F1-Score	Assessment
Naive Bayes	0.95	0.9530	Acceptable Baseline
Logistic Regression	0.99	0.9876	Strong Baseline
Linear SVM	0.99	0.9942	Best Performing

Table 3: Final Classification Report — Linear SVM

Class	Precision	Recall	F1-Score	Support
Fake (0)	0.99	0.99	0.99	4696
Real (1)	0.99	1.00	0.99	4242
Macro Avg	0.99	0.99	0.99	8938

The confusion matrix recorded only 56 false positives and 32 false negatives, confirming balanced, reliable classification. The trained model is serialised as a .pkl file containing the TF-IDF vocabulary, learned SVM weights, and optimal hyperparameters—ready for direct reuse in the Milestone 2 pipeline.

3 Milestone 2 — Objectives and Design Philosophy

3.1 Core Objectives

1. Accept article URLs or raw text and extract clean, structured content.
2. Produce a fast preliminary credibility signal via the Milestone 1 ML model.
3. Extract key verifiable claims from the article using an LLM.
4. Retrieve semantically similar historical fact-check claims from ChromaDB (Static RAG).
5. Fetch current, real-time news evidence related to extracted claims via NewsAPI (Dynamic RAG).
6. Apply LLM-based multi-source reasoning to generate an explainable credibility report.
7. Orchestrate all steps through a LangGraph five-node agent workflow with explicit state.
8. Expose functionality via FastAPI REST endpoints consumed by a Streamlit frontend.
9. Deploy publicly on Streamlit Community Cloud or Hugging Face Spaces.

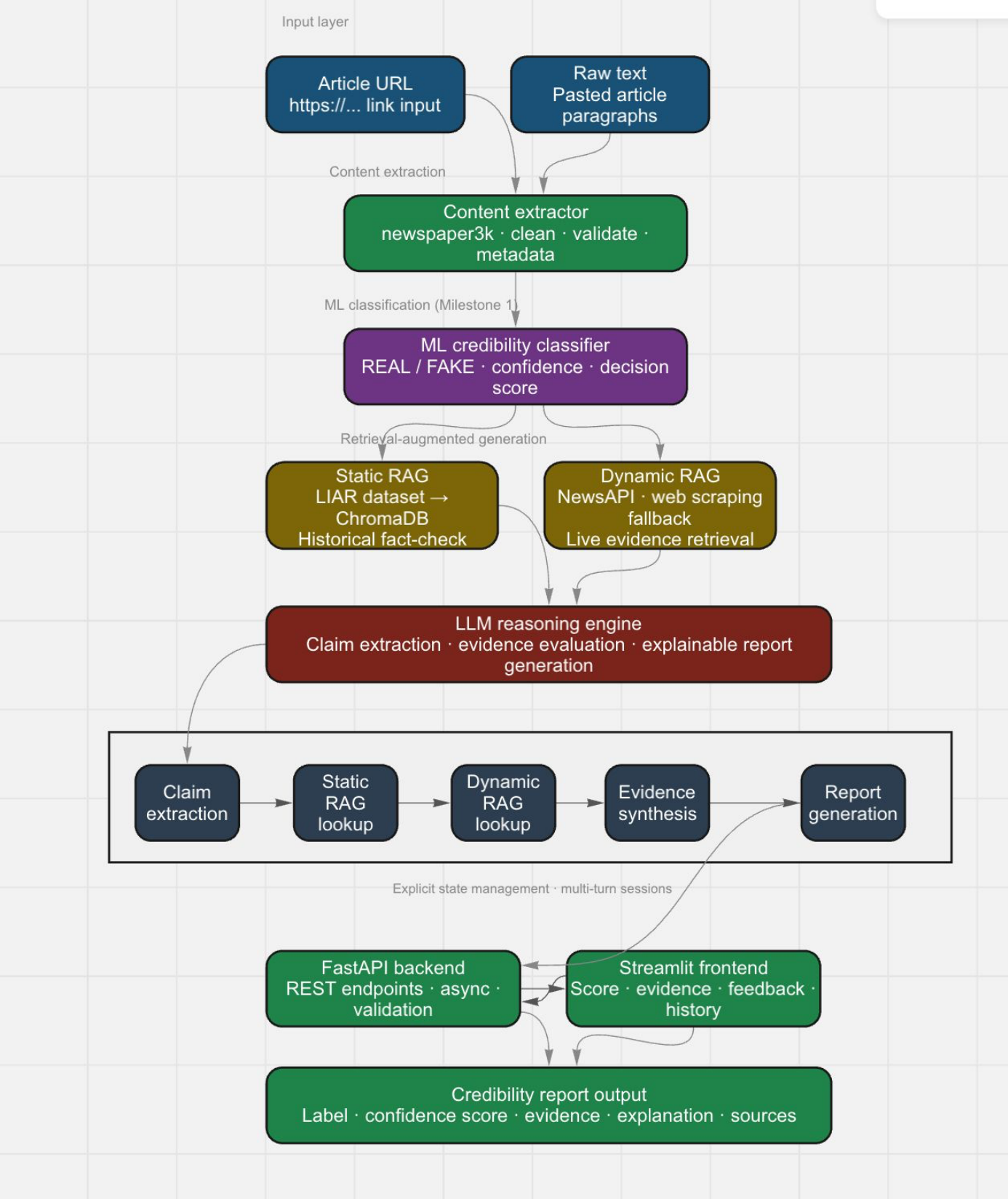
3.2 Design Philosophy

The system does not replace the ML classifier with an LLM. The ML model provides a fast, high-precision binary signal; the Static RAG layer provides grounded historical evidence; the Dynamic RAG layer adds real-time context; and the LLM synthesises all three into a coherent, explainable verdict. This **progressive trust pipeline** ensures every claim in the final report is anchored to retrieved evidence rather than generated from parametric memory alone.

4 System Architecture

4.1 Architecture Diagram

Figure 1 presents the complete system architecture, accurately reproducing the multi-layer hybrid pipeline designed by the team. Colour coding mirrors the original diagram: dark navy for input/agent nodes, teal green for extraction/output, purple for ML classification, golden olive for RAG components, and crims



on for LLM reasoning.

Figure 1: Milestone 2 — Complete Hybrid Intelligence System Architecture

4.2 Four-Layer Architecture Overview

Table 4: System Layer Decomposition

Layer	Components
Presentation Layer	Streamlit web UI; text/URL input; confidence gauge; evidence panel; structured report display; history tracking; conversational chatbot interface
API Layer	FastAPI REST endpoints; Pydantic request validation; async processing; standard JSON responses
Agent Layer	LangGraph five-node workflow; explicit state management; multi-turn session memory; reproducible reasoning
Intelligence Layer	Linear SVM ML classifier + Static RAG (ChromaDB/LIAR) + Dynamic RAG (NewsAPI/web scraping) + LLM reasoning engine

5 Detailed Workflow Description

The proposed system follows a hybrid, multi-stage workflow that integrates traditional machine learning with modern agentic AI techniques to perform comprehensive news credibility analysis. Each stage is described in detail below.

5.1 Stage 1 — User Input and Content Extraction

The process begins with user input, where the system accepts either raw news article text or a URL. If a URL is provided, the content is extracted and preprocessed using newspaper3k, which scrapes the webpage, isolates the article body, and removes HTML artefacts. Standard NLP preprocessing is then applied: tokenisation, lowercasing, removal of special characters, stop-word filtering with negation preservation, and lemmatisation. The output is clean, normalised article text accompanied by structured metadata.

ContentExtractor—OutputSchema	
success	true/false
content	Cleaned,normalisedarticletext
metadata.source	"url" or "raw text"
metadata.title	Extractedarticletitle
metadata.author	Extractedauthor(s)
metadata.publish date	Extractedpublicationdate(ifavailable)
metadata.word count	Wordcountofcleanedtext

5.2 Stage 2 — ML Credibility Classification

The cleaned text is passed through the Milestone 1 ML inference pipeline. The TF-IDF vectoriser transforms the text into a high-dimensional sparse feature vector, which the optimised Linear SVM classifies. This stage produces a binary **label** (REAL or FAKE), a **confidence score** $\in [0,1]$, and a raw **decision score** representing the signed distance from the hyperplane. This score serves as a *preliminary signal* rather than a final decision, informing but not overriding the subsequent evidence-based reasoning.

5.3 Stage 3 — Claim Extraction via LLM

Upon entering the agentic pipeline orchestrated by LangGraph, the first node invokes the

LLM to extract key verifiable claims from the article text. Rather than reasoning over the entire article, the agent focuses analytical effort on the specific statements most likely to be misleading or factually incorrect. Claim extraction is performed using structured chain-of-thought prompting with explicit output format requirements to ensure parseable, downstream-compatible results.

5.4 Stage 4 — Static RAG: Historical Fact-Check Retrieval

The extracted claims are passed to the Static RAG component, which queries a **ChromaDB** vector database populated with the LIAR dataset—containing over 12,000 politically oriented claims with human-assigned credibility labels and contextual justifications. Semantic similarity search retrieves the top- k most relevant historical claims along with their verified verdicts and explanations, providing an evidence base grounded in previously validated information.

Static RAG Technical Configuration

Embedding Model	Sentence transformer (all-MiniLM-L6-v2)
Vector Store	ChromaDB (persistent collection)
Retrieval	Top- k cosine similarity, $k = 5$ by default
Output per hit	{claim, credibility label, explanation, source}

5.5 Stage 5 — Dynamic RAG: Real-Time Evidence Retrieval

To address the recency limitation of static datasets, the agent invokes external tools to fetch current, real-time evidence. **NewsAPI** integration sends queries derived from extracted claims to retrieve recent articles, ranked by relevance and recency. A **web scraping fallback** handles sources not indexed by NewsAPI. Retrieved documents undergo deduplication, relevance scoring, and noise reduction before being passed to the reasoning engine.

At the conclusion of Stage 5, the LangGraph agent holds three complementary information sources: the ML prediction, historical RAG evidence, and live news evidence.

5.6 Stage 6 — LLM Multi-Source Reasoning and Report Generation

The final agentic step involves comprehensive multi-source reasoning. The LLM synthesises all available inputs through a structured chain: (1) each extracted claim is evaluated against historical and real-time evidence; (2) cross-source consistency is assessed for agreements and contradictions; (3) risk factors are identified—sensational language, missing citations, logical inconsistencies; and (4) a structured credibility report is generated per the schema in Section 6.

Prompting strategies minimise hallucination through evidence-grounded generation (the LLM must cite retrieved evidence for every claim), chain-of-thought reasoning, and lowtemperature decoding for factual consistency.

5.7 Stage 7 — Conversational Chatbot Interface

Beyond single-shot analysis, the system supports a **conversational chatbot interface** for follow-up questions about the analysed article. This chatbot leverages the same RAG and tool-based retrieval mechanisms along with conversational memory maintained through LangGraph session state, enabling contextual, multi-turn responses.

6 Output Specification

Summary	Article overview with extracted key claims
Risk Factors	Sensational language, missing citations, inconsistencies detected
ML Signal	Binary label (REAL/FAKE) with confidence score
Static Evidence	Top- <i>k</i> LIAR dataset claims with credibility labels and explanations
Live Evidence	Related current articles from NewsAPI with source links
Cross-Source	LLM reasoning over claim–evidence consistency across sources
Verdict	Final credibility: High / Low / Uncertain
Confidence	Aggregated signal from ML + RAG + LLM reasoning
Source Links	Clickable URLs for transparency and user verification
Disclaimer	Ethical AI notice: automated assessments are advisory, not definitive

7 FastAPI Backend and Streamlit Frontend

7.1 FastAPI Endpoint Design

Table 6: Planned FastAPI REST Endpoints

Method	Endpoint	Description
POST	/analyze	Submit URL or raw text; returns full credibility report
POST	/predict	ML-only fast prediction; label and confidence only
POST	/chat	Conversational queries with LangGraph session memory
GET	/health	Service health check and model load status
GET	/history	Retrieve past analysis results by session ID

All endpoints implement Pydantic request validation, return standard JSON responses, and support asynchronous processing for long-running agent tasks.

7.2 Streamlit Frontend Features

- Article text area and URL input with auto-detection of input type
- Real-time credibility score with visual confidence gauge
- Expandable evidence panel showing retrieved fact-check claims and sources
- Structured report display with risk factors, analysis, and final verdict
- Multi-turn chatbot interface for follow-up questions with session memory
- History tracking and user feedback submission for continuous improvement

8 Limitations and Risk Mitigation

Table 8: Known Limitations and Mitigation Strategies

Limitation	Mitigation Strategy
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ML fidelity	domain	speci-	ISOT model may not generalise to all domains; ML signal treated as preliminary, never final decision
NewsAPI rate limits			Response caching; scraping fallback; request throttling
LLM hallucination			Evidence-grounded prompting; explicit citation requirement; chain-of-thought; low temperature
LIAR cency	dataset	re-	Dynamic RAG compensates with live evidence; periodic ChromaDB refresh planned post-launch
Open-source LLM latency			Lightweight models via Groq free tier; async FastAPI prevents UI blocking
Text-only scope			Image and video credibility noted as future work

9 Conclusion

Milestone 2 represents a substantial architectural and capability advancement over Milestone 1. The system evolves from a single-model binary classifier into a fully agentic, multi-source misinformation monitoring platform that produces structured, evidence-backed credibility reports. The three completed phases provide a tested foundation: reliable content ingestion, an integrated ML prediction layer with 99% F1-score, and a clean modular project structure. The six planned phases build progressively on this base, introducing retrieval-augmented generation, LLM-based reasoning, LangGraph agent orchestration, and full-stack deployment.

The hybrid design philosophy ensures the system is simultaneously *accurate* (leveraging the high-precision ML signal), *grounded* (anchoring every verdict in retrieved historical and real-time evidence), and *explainable* (producing structured reports that justify the final verdict to the end user). Upon completion, this system will represent a robust, responsible, and scalable solution to automated, transparent news credibility assessment—a genuinely meaningful contribution to media integrity in the age of digital misinformation.

10 References

- [1] Ahmed, H., Traore, I., & Saad, S. (2018). Detection of Online Fake News Using N-gram Analysis and Machine Learning Techniques. *ISDDC 2018*.
- [2] ISOT Fake News Dataset. University of Victoria. <https://www.kaggle.com/datasets/clmentbisailon/fake-and-real-news-dataset>
- [3] LangGraph Documentation. LangChain Inc. <https://langchain-ai.github.io/langgraph/>
- [4] ChromaDB Documentation. Chroma AI. <https://docs.trychroma.com/>
- [5] NewsAPI Documentation. <https://newsapi.org/docs>